

Indian Institute of Technology Madras

DA5400W - Foundations of Machine Learning

Quiz II Solutions

Date: March 22, 2026

Duration: 1.5 hours — Total Marks : 20

Questions and Solutions

1. **(2 Marks)** Which of the following equations can be reformulated as linear regression models, treating (x, y) as variables and (a, b) as parameters? For each valid case, show the algebraic transformation that converts the equation into linear regression form.

(a) $y = ae^{b/x}$

(b) $y = \frac{ax + b}{x + 1}$

(c) $y = a \ln(x + b)$

(d) $y = \frac{a}{1 + \exp^{-a(x-b)}}$

Solution

(a) $y = ae^{b/x}$ is linearisable (after log-transform).

Assume $y > 0$ (so $\ln y$ is defined) and $x \neq 0$.

$$y = ae^{b/x} \Rightarrow \ln y = \ln a + \frac{b}{x}.$$

Let

$$Y := \ln y, \quad \phi(x) := \frac{1}{x}, \quad \alpha := \ln a.$$

Then

$$Y = \alpha + b\phi(x),$$

(b) $y = \frac{ax + b}{x + 1}$ is linearisable (by transforming the response).

Assume $x \neq -1$.

$$y = \frac{ax + b}{x + 1} \Rightarrow y(x + 1) = ax + b \Rightarrow yx + y = ax + b.$$

Define a transformed response

$$Y := y(x + 1).$$

Then the model becomes

$$Y = ax + b,$$

(c) $y = a \ln(x + b)$ is not linearisable in (a, b) .

(d) $y = \frac{a}{1 + \exp^{-a(x-b)}}$ is not linearisable in (a, b) .

2. **(4 Marks)** The purity of oxygen produced by a fractionation process is thought to be related to the percentage of hydrocarbons (HC) in the main condenser of the processing unit according to the simple linear regression model

$$y_i = a + bx_i + \epsilon_i, \quad i = 1, 2, \dots, n,$$

where y_i denotes the oxygen purity when the HC percentage is x_i , and the error terms ϵ_i are independent with mean zero and variance σ_ϵ^2 .

A total of $n = 40$ experiments were conducted, and the data are summarized as:

$$\begin{aligned} \sum_{i=1}^n x_i &= 20, & \sum_{i=1}^n y_i &= 38, \\ \sum_{i=1}^n x_i^2 &= 17, & \sum_{i=1}^n x_i y_i &= 40.7, \end{aligned}$$

Compute the OLS point estimates of the intercept a and the slope b .

Hint: Use the fact that at the optimum,

$$\sum_{i=1}^n (y_i - a - bx_i) = 0 \quad \text{and} \quad \sum_{i=1}^n x_i (y_i - a - bx_i) = 0.$$

Solution

Given the simple linear regression model

$$y_i = a + bx_i + \epsilon_i, \quad i = 1, \dots, n,$$

with summary statistics (for $n = 40$):

$$\sum_{i=1}^n x_i = 20, \quad \sum_{i=1}^n y_i = 38, \quad \sum_{i=1}^n x_i^2 = 17, \quad \sum_{i=1}^n x_i y_i = 40.7$$

First compute the sample means:

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i = \frac{20}{40} = 0.5, \quad \bar{y} = \frac{1}{n} \sum_{i=1}^n y_i = \frac{38}{40} = 0.95$$

Compute the centered sums:

$$S_{xx} = \sum_{i=1}^n (x_i - \bar{x})^2 = \sum_{i=1}^n x_i^2 - n\bar{x}^2 = 17 - 40(0.5)^2 = 17 - 10 = 7,$$

$$S_{xy} = \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}) = \sum_{i=1}^n x_i y_i - n\bar{x}\bar{y} = 40.7 - 40(0.5)(0.95) = 40.7 - 19 = 21.7$$

Hence the OLS slope and intercept are:

$$\hat{b} = \frac{S_{xy}}{S_{xx}} = \frac{21.7}{7} = 3.1,$$

$$\hat{a} = \bar{y} - \hat{b}\bar{x} = 0.95 - (3.1)(0.5) = 0.95 - 1.55 = -0.6$$

$\hat{a} = -0.6, \quad \hat{b} = 3.1$

3. You are given a simple regression setting with a single feature x and response y . Assume we fit a no-intercept model on a training set of $n = 4$ points:

$$(x, y) = \{(1, 2), (2, 3), (3, 2), (4, 5)\}.$$

Consider the ridge-regularized least squares objective (one parameter w):

$$J(w) = \sum_{i=1}^4 (y_i - wx_i)^2 + \lambda w^2, \quad \lambda \geq 0.$$

- (a) **(2 Marks)** Which of the following gives the ridge solution $\hat{w}(\lambda)$.

- i. $\hat{w}(\lambda) = \frac{\sum x_i y_i}{\sum \lambda x_i^2}$
- ii. $\hat{w}(\lambda) = \frac{\sum -x_i (y_i - wx_i)}{\lambda}$
- iii. $\hat{w}(\lambda) = \frac{\sum x_i y_i}{\lambda + \sum x_i^2}$
- iv. $\hat{w}(\lambda) = \frac{\sum x_i y_i}{\sum (x_i^2 + \lambda y_i)}$

Answer: Option (iii)

- (b) **(2 Marks)** Compute $\hat{w}(\lambda)$ for $\lambda = 1$

Solution

Expand the objective:

$$J(w) = \sum (y_i^2 - 2wx_i y_i + w^2 x_i^2) + \lambda w^2.$$

Differentiate with respect to w :

$$\frac{dJ}{dw} = -2 \sum x_i y_i + 2w \sum x_i^2 + 2\lambda w.$$

Set derivative to zero:

$$-2 \sum x_i y_i + 2w \left(\sum x_i^2 + \lambda \right) = 0.$$

Divide by 2 and rearrange:

$$w \left(\sum x_i^2 + \lambda \right) = \sum x_i y_i.$$

$$\boxed{\hat{w}(\lambda) = \frac{\sum x_i y_i}{\sum x_i^2 + \lambda}}$$

Numerical Computation

Compute required sums:

$$\sum x_i^2 = 1^2 + 2^2 + 3^2 + 4^2 = 30,$$

$$\sum x_i y_i = 1 \cdot 2 + 2 \cdot 3 + 3 \cdot 2 + 4 \cdot 5 = 34$$

Hence,

$$\hat{w}(\lambda) = \frac{34}{30 + \lambda}$$

For $\lambda = 1$:

$$\hat{w}(1) = \frac{34}{31} = 1.097$$

$$\boxed{\hat{w}(1) = 1.097}$$

4. **(1 Mark)** For a linear regression model build on a dataset, it is observed that R^2 value on the training set is found to be negative. Which of following statement is true in this scenario?
- (a) Model partly captures the variance in the dependent variable
 - (b) Variance in the dependent variable is too high and cannot be captured by the model
 - (c) Mean value of the dependent variable is better prediction than the predictions made by this model
 - (d) Scenario is not possible as R^2 cannot be negative

Answer: Option (c)

5. **(2 Marks)** A researcher models the relationship between weekly exercise hours (x) and the **final body weight** (y) of participants using the log-transformed linear regression model

$$\log(y) = a - 0.2x.$$

Which of the following statements is correct?

- (a) For every additional hour of exercise, the final body weight decreases by exactly 0.2 kilograms.
- (b) For every additional hour of exercise, the final body weight becomes approximately 18% lower.
- (c) For every additional hour of exercise, the final body weight becomes exactly 0.2% lower.
- (d) For every additional hour of exercise, the logarithm of the final body weight approximately decreases by 2%.

Answer: Option (b)

Explanation: Given the regression model

$$\log(y) = a - 0.2x,$$

a one-unit increase in x changes $\log(y)$ by -0.2 , so

$$\log(y_{\text{new}}) - \log(y_{\text{old}}) = -0.2.$$

Using log rules,

$$\log\left(\frac{y_{\text{new}}}{y_{\text{old}}}\right) = -0.2.$$

Exponentiating both sides gives

$$\frac{y_{\text{new}}}{y_{\text{old}}} = e^{-0.2}.$$

Now,

$$e^{-0.2} \approx 0.8187.$$

Thus the decrease is

$$1 - 0.8187 = 0.1813 \approx 18\%.$$

6. **(2 Marks)** In a logistic regression model with one feature x , the coefficient of x is positive. Which of the following is true?

- (a) As x increases, the probability of class 1 always decreases
- (b) As x increases, the log-odds of class 1 increase
- (c) The relationship between x and class probability is linear
- (d) A positive coefficient implies zero classification error

Answer: Option **(b)**

7. **(1 Mark)** Why is Naive Bayes called “naive”?

- (a) It uses no probability model
- (b) It assumes all features are conditionally independent given the class
- (c) It only works for Gaussian data
- (d) It ignores class priors

Answer: Option **(b)**

8. **(1 Mark)** Which of the following statements is generally true as k is increased in a KNN classifier?

- (a) Variance tends to increase and bias tends to decrease
- (b) Variance tends to decrease and bias tends to increase
- (c) Both variance and bias decrease
- (d) Both variance and bias increase

Answer: Option **(b)**

Explanation: In a KNN classifier, the parameter (k) determines how many nearby training points are used to classify a new observation.

When (k) is small, the classifier looks at only a few neighbors. This makes the model very flexible, so it can adapt closely to the training data. However, it also means the prediction can change a lot if the training data changes slightly. Therefore, the model has:

[low bias, high variance.]

When (k) is increased, the classifier averages over more neighbors. This makes the decision boundary smoother and less sensitive to small changes in the training data. So variance decreases.

However, using many neighbors makes the classifier less flexible. It may smooth over important local patterns and fail to capture the true structure of the data. So bias increases.

9. **(2 Marks)** A Bernoulli Naive Bayes classifier is trained with two binary features x_1, x_2 and two classes C_1, C_2 .

Class priors:

$$P(C_1) = 0.7, \quad P(C_2) = 0.3$$

Feature likelihoods:

$$P(x_1 = 1 \mid C_1) = 0.8, \quad P(x_2 = 1 \mid C_1) = 0.4$$

$$P(x_1 = 1 \mid C_2) = 0.3, \quad P(x_2 = 1 \mid C_2) = 0.9$$

For a test point with

$$x_1 = 1, \quad x_2 = 0$$

compute the unnormalized posterior score for Class 1.

Solution: Given:

$$P(C_1) = 0.7$$

$$P(x_1 = 1 \mid C_1) = 0.8, \quad P(x_2 = 1 \mid C_1) = 0.4$$

For the test point:

$$x_1 = 1, \quad x_2 = 0$$

Since this is Bernoulli Naive Bayes, we also need

$$P(x_2 = 0 \mid C_1) = 1 - 0.4 = 0.6$$

The unnormalized posterior score for Class 1 without log is

$$P(C_1) P(x_1 = 1 \mid C_1) P(x_2 = 0 \mid C_1)$$

Substituting values:

$$= 0.7 \times 0.8 \times 0.6$$

$$= 0.336$$

Using natural logarithm:

$$= \ln(0.7) + \ln(0.8) + \ln(0.6)$$

$$= -0.3567 + (-0.2231) + (-0.5108)$$

$$= -1.0906$$

Using base-10 logarithm:

$$= \log_{10}(0.7) + \log_{10}(0.8) + \log_{10}(0.6)$$

$$= -0.1549 + (-0.0969) + (-0.2218)$$

$$= -0.4736$$

10. **(1 Mark)** Which of the following statements best describes the Total Least Squares (TLS) approach?
- (a) TLS assumes that only the predicted variable contains measurement errors.
 - (b) TLS minimizes the vertical distances between the data points and the fitted line.
 - (c) TLS accounts for errors in both the input variables and the predicted variable.
 - (d) TLS can be used only when there is exactly one input variable.

Answer: Option (c)